Integration of PM-runtime with System-wide Power Management

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September 10, 2019

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It Is All About Energy-efficiency

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Two Different Ways To Get There

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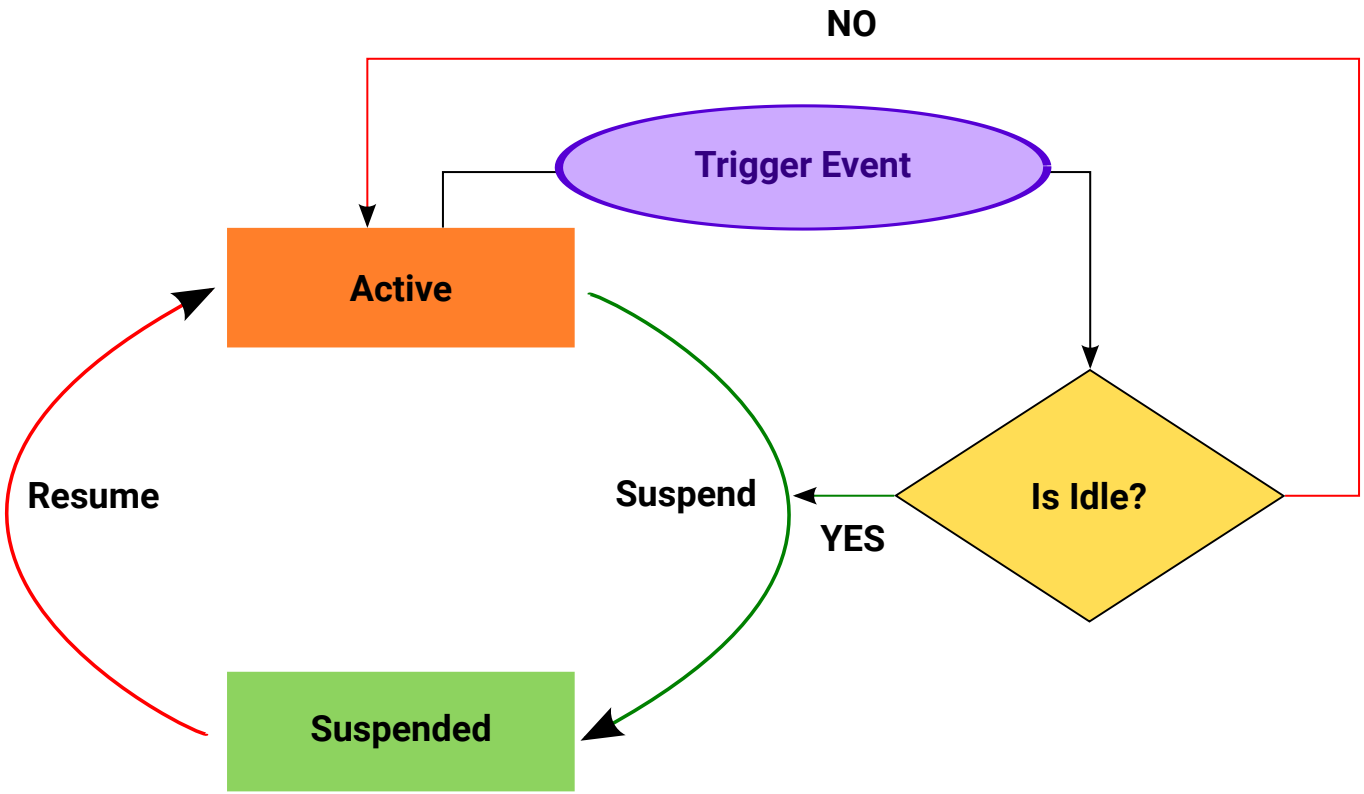
High-level View to PM-runtime



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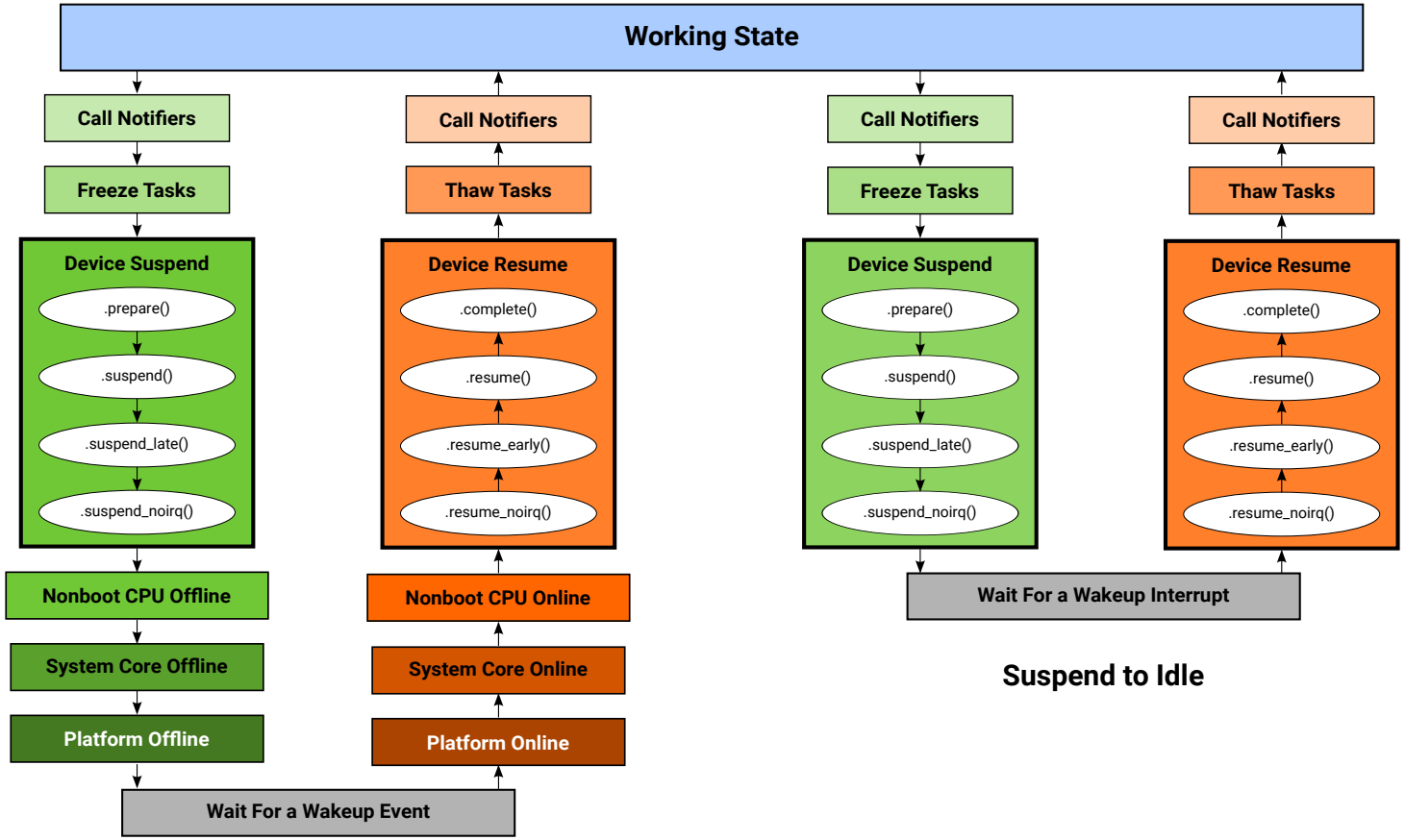
System Suspend Control Flows



Platform-based Suspend

Suspend to Idle

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Transparent Handling of Suspended Devices

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PM-runtime Callbacks Re-Use

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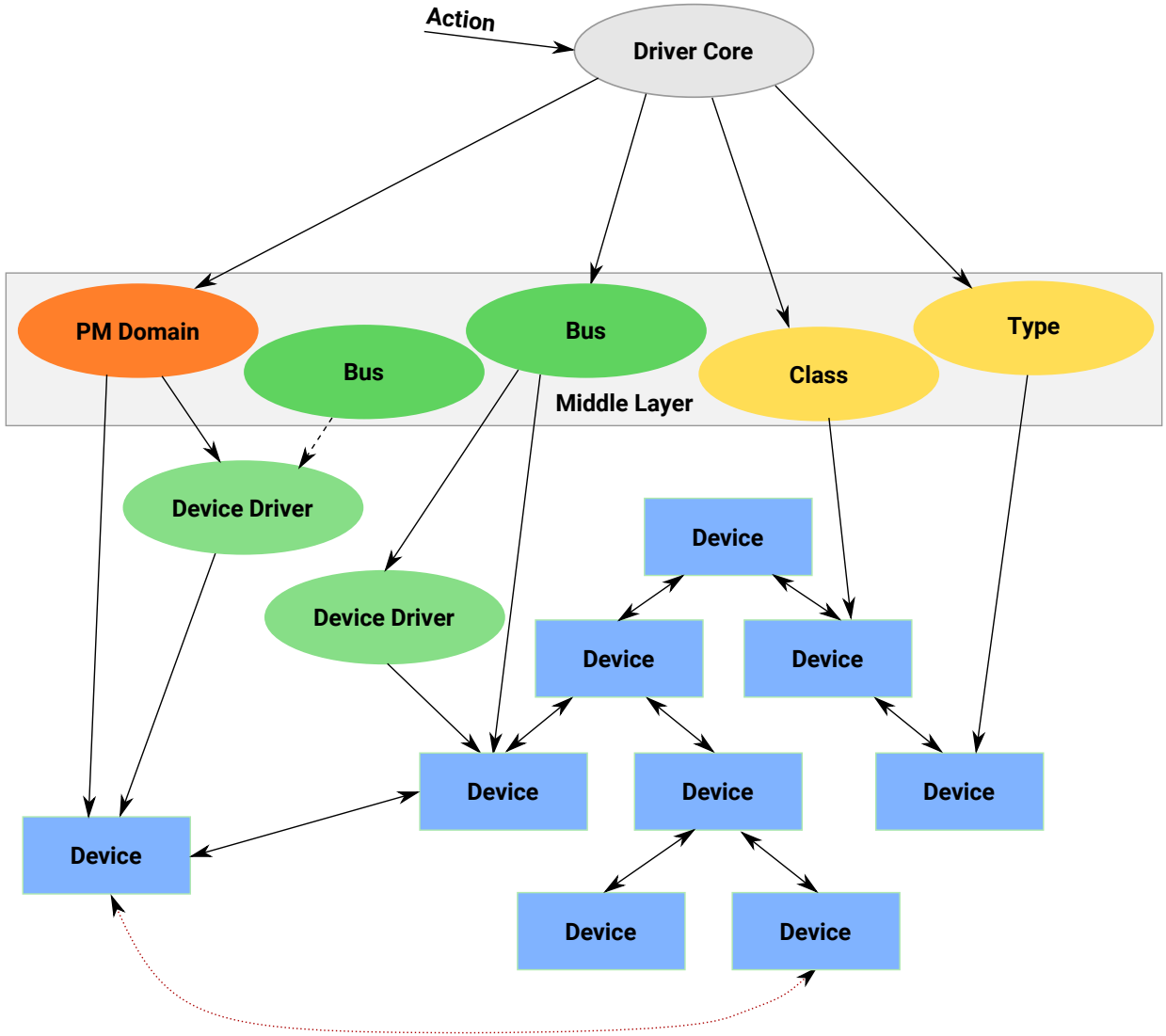
Driver Core and Power Management



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Devices May Need To Be Reconfigured

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Differences Related To Wakeup

PM-runtime

Device wakeup always enabled.

System-wide PM

`/sys/devices/.../power/wakeup`: enabled or disabled.

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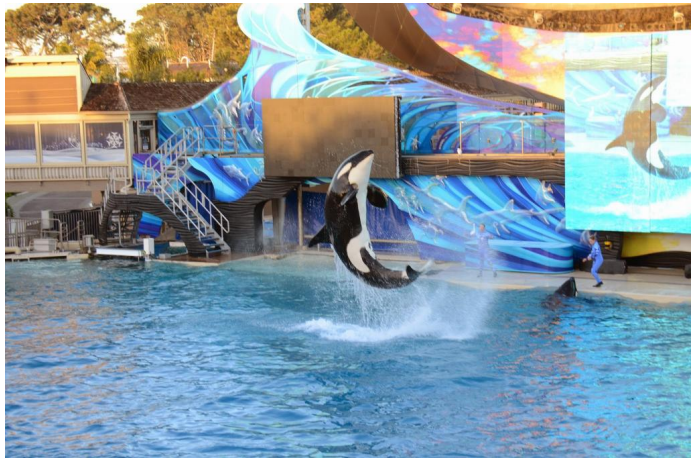
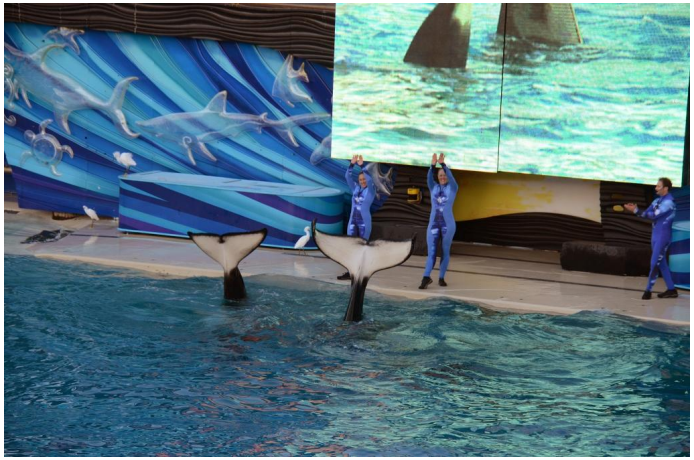
Intermittent Interactions Between Devices

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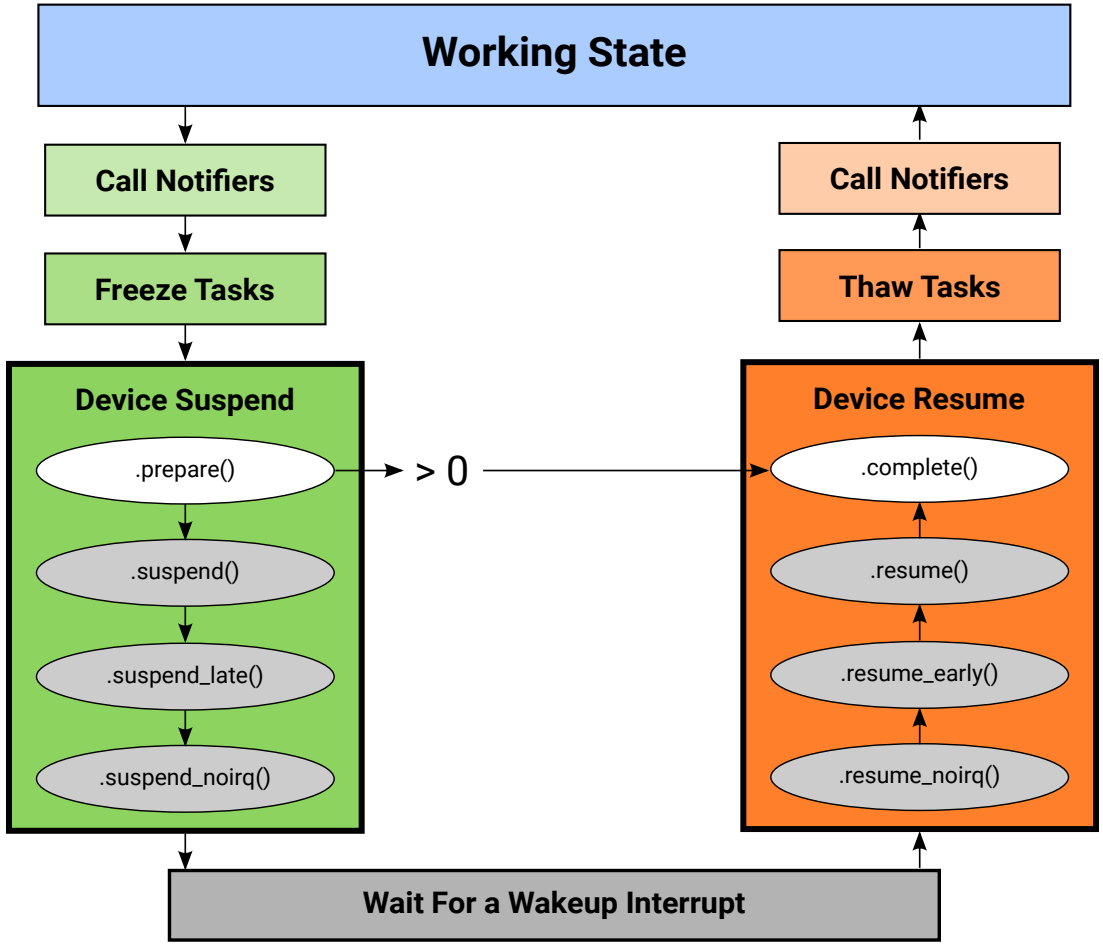
Direct-complete Optimization Idea



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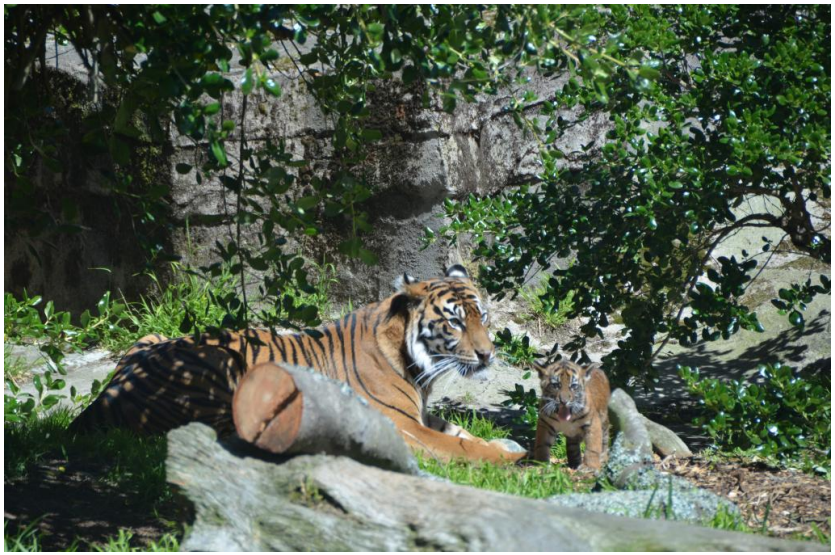
Limitations Of Direct-complete

1. Devices subject to intermittent interactions cannot do it.
2. Cannot be done if the children do not do it too.



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Wrappers Around PM-runtime Callbacks

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Limitations Of Callback Wrappers

1. Invoke middle-layer PM-runtime callbacks.
2. Disable PM-runtime.
3. Questionable approach to leaving devices in suspend.

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Driver Flags For System-wide Power Management

- DPM_FLAG_NEVER_SKIP
- DPM_FLAG_SMART_PREPARE
- DPM_FLAG_SMART_SUSPEND
- DPM_FLAG_LEAVE_SUSPENDED

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Coverage Gaps

1. Runtime resume resulting from intermittent interactions not covered.
2. `DPM_FLAG_SMART_SUSPEND` (generally) required for callbac re-use.

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PM-runtime And System-wide PM: Observations

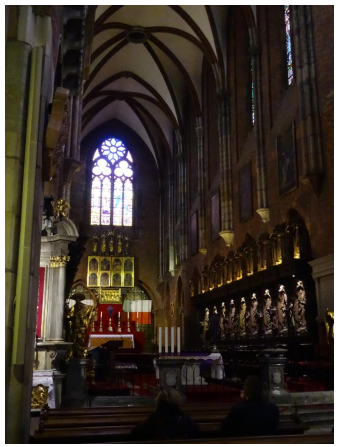
Observation 1

Devices cannot be runtime-suspended during system-wide suspend/resume.

Observation 2

Two cases for runtime resume during system-wide PM transitions:

- Called from system-wide PM callbacks for the same device.
- Called from somewhere else.



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Runtime Resume During System-wide Suspend

1. If called from system-wide PM callbacks for the same device, run it.
2. Otherwise:
   - If called after some system-wide PM code has run, block it.
   - Else (always) defer suspending the target device to the “noirq” phase.

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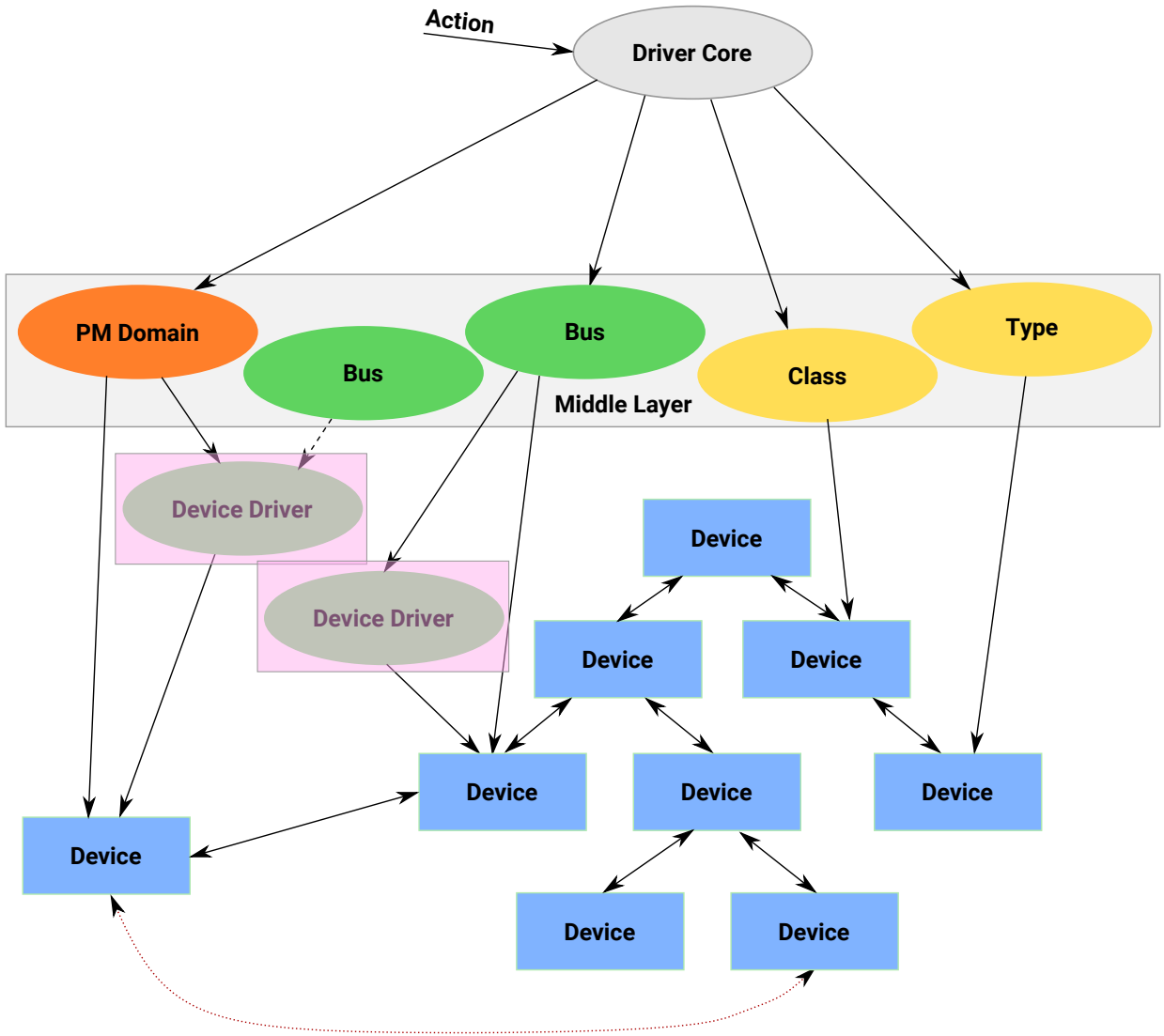
Generally, Callback Wrappers Do Not Help



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Plan: Eliminate The Reason For Using Callback Wrappers

Concern addressed by them

Prevent the same callback from running twice in a row back to back.

There are cases in which they are not necessary already

- PM-aware bus types (PCI, USB).
- ACPI PM domain.

What can be done elsewhere

1. Make all PM-aware middle-layer code honor PM-runtime status.
2. For pass-through middle layers, make the PM core do that.

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Comments, Questions, Concerns?

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References

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